

many of the concepts and licensing conventions developed over a decade by software pioneers.

Recently, open source silicon platforms have begun to emerge as the third phase of the movement, which is re-democratising innovation and powering the global tech economy.

A tale of two innovations

Phase 1: Linux. One of open source software's biggest successes is Linux operating system (OS), conceived by Linus Torvalds as a free alternative to AT&T's powerful but costly UNIX OS. Fittingly, Linux borrows many elements from Richard Stallman's earlier GNU project. Within a few years of its initial release in 1991, Linux had attracted a large, active community of users, many of whom began contributing to its vast and growing pool of user-created resources.

As a result, Linux developers can access countless libraries of pre-written utilities and functional blocks, created by fellow users. With many of the low-level elements of their applications already available, developers are able to focus their efforts on the proprietary functions that add value to their product and differentiate it from its competitors. Linux's ability to shorten a product's time-to-market while reducing development costs made it a top choice for embedded systems that run under the hood of thousands of consumer and industrial products. In addition, Linux powers the servers that run 96.5 per cent of the top one million domains in the world (as ranked by Alexa), 92 per cent of Amazon's 350,000 EC2 Web servers and all 500 of the world's most powerful supercomputers.

Phase 2: Arduino. Arduino embedded computing platform was at the vanguard of the second open source wave (hardware) when it was first conceived in 2003 as a student project at Interaction Design Institute Ivrea in Ivrea, Italy. The project's audacious goal was to develop a very-low-cost, easy-to-program, single-board computer that would enable designers, artists and other non-engineers to build digital devices and interactive objects that can sense and control objects in the physical and

Open standards versus open source

Open standards helped make our industrial economy possible. These do not tell a manufacturer anything about how to design or make a product; these define its basic functionality, especially with regards to how it interacts with similar products from other manufacturers. At the dawn of the Industrial Revolution, standards for things like thread pitches of bolts and screws, calibre of bullets, and railway cars' wheel spacing made these easier and less expensive to manufacture, thereby creating a broader demand for them.

In more recent times, open standards have allowed products ranging from cellphones and wireless LAN components, to batteries and computer memories enjoy the benefits arising from The Network Affect—a phenomenon in which value of a product or service increases according to the number of others using it (Carl Shapiro and Hal R. Varian, 1999, in their book titled *Information Rules*). In fact, it is difficult to imagine these enjoying such widespread success had these been based on a patchwork of proprietary, non-interoperable technologies.

Open source technologies expand on this concept by providing detailed information about the functionality of a hardware or software element, and most or all of the information needed to implement it. Much like how Wikipedia relies on its users to keep its entries accurate and up-to-date, most open source software platforms rely on their user communities to develop applications, bug fixes, and improved versions of the original creation and its associated support tools. For hardware designs, this support also includes mechanical drawings, schematics, firmware, component and material lists, and assembly instructions.

The Network Effect also helps drive the success of open source technologies. Once a platform's user community reaches a certain level, it forms a so-called virtuous circle, whereby the abundance of development resources attracts even more users. In the case of hardware platforms, the growing user base stimulates the emergence of an ecosystem of vendors who compete to offer components, assemblies and accessories at increasingly attractive prices.

digital world. In retrospect, developers significantly underestimated the project's actual impact.

The original Arduino board was roughly the size of a business card and provided easy access to signal lines of its 8-bit microcontroller (MCU) through a series of pins along each side of the board. These pins allow a developer to easily connect the processor's input and output lines to a real-world device, such as a light sensor or motor. These can also be used to connect drop-in daughter boards (known as shields) equipped with sensors, network interfaces or other functionalities. Programming the board is accomplished using simple, English-like commands that are written and compiled within a user-friendly integrated development environment (IDE).

Arduino's low cost, versatility and ease-of-use led to its explosive popularity with technology hobbyists who embraced the do-it-yourself (DIY) philosophy of the so-called maker movement. Sold in millions, the tiny open source board and its variants served as the silicon intelligence that enabled hobbyists to unleash their imaginations and create things like low-cost automated telescopes, smart clothing, personal robots and automated home brewing systems.

Meanwhile, as enterprising makers

began to offer innovative commercial products on Kickstarter and other crowdfunding sites, Arduino technology helped them bring their ideas to market quickly and inexpensively.

A powerful legacy. As some of the earliest open source platforms, Linux and Arduino were instrumental in upending the conventional top-down business model in which proprietary technologies were developed in isolation by large corporations and doled out to those who could afford them. In contrast, open source's bottom-up approach enables designers to harness the collective intelligence of a user community to create tools and products that reflect real-world needs.

The overwhelming success of Linux and Arduino also demonstrated that product differentiation on a common, non-proprietary platform is not only possible but desirable.

Open source technologies perform another vital and surprising role within the tech economy by serving as a catalyst that enables the creation of other open source technologies that make new and unexpected applications possible. In much the same way that Stallman's GNU project provided many of the concepts and enabling technologies that made Linux possible, Arduino gave makers, tinkers and would-be entrepreneurs the low-cost, easily-hackable controller